

SMART INDIA HACKATHON – IDEA TEMPLATE

Project: Agro Vision

Problem Statement ID	SIH26131
Organization	Government of Maharashtra
Problem Statement	Early detection and management of crop diseases and pest infestations
Theme	Agriculture, FoodTech & Rural Development
Category	Software
Proposed Solution Name	Agro Vision

1. Problem Description

Farmers may face crop diseases and pest infestations that can spread rapidly and reduce crop quality and yield. Early identification can help farmers take timely management measures.

2. Proposed Solution

Agro Vision is a software-based intelligent crop monitoring and disease-management platform. The system allows users to provide crop images and receive disease/pest identification support, along with relevant management guidance. The platform can also maintain records and present useful information through a simple farmer-oriented interface.

3. Key Features

- Crop image upload and analysis
- Disease/pest identification using an AI/ML model
- Disease information and recommended management measures
- Farmer-friendly dashboard
- Crop/disease history and record management
- Alerts or notifications for identified problems
- Admin/model-data management support

4. Innovation / Novelty

The proposed system combines image-based crop health analysis with a simple digital workflow for detection, information delivery and management. The focus is on making AI-assisted crop monitoring accessible through a straightforward software interface.

5. Technology Stack

Frontend: HTML, CSS, JavaScript / responsive web interface

Backend: Python with Flask

AI/ML: Python-based machine learning/deep learning image-classification model

Database: SQLite / SQL database

Data Processing: Python, NumPy, Pandas and image-processing libraries as required

Deployment: Cloud/server deployment as required

6. Expected Impact

Agro Vision is intended to support earlier awareness of crop diseases and pest infestations, help users understand the detected issue, and assist in taking timely management action. The solution can also support digital record keeping and scalable AI-assisted crop monitoring.

7. Feasibility

The proposed solution can be implemented using commonly available software technologies and publicly available agricultural image datasets. A modular architecture allows the image-analysis model, backend, database and user interface to be developed and tested independently before integration.

8. Implementation Plan

1. Collect and prepare crop disease/pest image data.
2. Preprocess and label the dataset.
3. Train and evaluate the AI/ML model.
4. Develop the Flask backend and database.
5. Develop the farmer-oriented web interface.
6. Integrate image prediction and management guidance.
7. Test the complete system and improve accuracy/usability.
8. Deploy the application for demonstration.

9. Future Scope

Future versions may include multilingual support, mobile application support, weather-based risk information, location-aware alerts, offline functionality, additional crops and diseases, and integration with IoT-based field sensors.